Predictive Modeling in Health: Research Protocol and Reproducibility

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Abstract. *This document is an initial research protocol for CMP263, Machine Learning, class A, 2026/2. It organizes dataset selection, exploratory analysis, predictive modeling, evaluation, interpretation, and reproducibility. The authors have not selected a dataset or conducted experiments. The research question, methods, and final abstract will be revised after joint approval and empirical validation. No findings are reported here.*

Draft status: planning scaffold; not a completed empirical paper.

1. Introduction

This project will critically evaluate predictive models for a health-related task. The dataset and research question remain pending joint approval. The final text must motivate a specific question beyond algorithm accuracy. The work belongs exclusively to CMP263, Machine Learning, class A, 2026/2, taught by Mariana Recamonde Mendoza at UFRGS.

2. Related Work

The review will examine prior prediction targets, eligibility criteria, validation strategies, and limitations for the selected data. The authors will explain how their question differs. Novelty has not been established; basic algorithm descriptions will remain concise.

3. Data and Study Population

Two Brazilian sources are initial candidates: the SIVEP-Gripe severe acute respiratory syndrome records [Ministério da Saúde 2026] and the 2019 National Health Survey microdata [Instituto Brasileiro de Geografia e Estatística 2022]. Neither source has been selected or downloaded. The final paper will retain references to the selected source and report the download URL, snapshot date, license or terms, dictionary, eligibility criteria, sample size, target distribution, exclusions, and missing values. The study unit and the availability of predictors at the stated prediction time must be documented before training.

4. Methods and Experimental Design

The final protocol will define the prediction target, population, estimand, primary metric, and evaluation scheme before model selection. Candidate classification algorithms are logistic regression, nearest neighbors, decision trees, random forests, and support vector machines, alongside a dummy baseline. Regression would require appropriate alternative estimators.

4.1. Preprocessing and Validation

Imputation, scaling, encoding, feature selection, and any resampling will be fitted only on the training portion of each split. All algorithms will share the same evaluation splits. Grouped or temporal data will require corresponding validation constraints. The final test data will be reserved until hyperparameters, calibration, and decision thresholds are fixed.

4.2. Interpretation and Reproducibility

The selected model will be interpreted with a suitable method and compared with an interpretable baseline. Predictive associations will not be treated as causal effects. Seeds, environment versions, data hashes, configurations, and execution instructions will accompany the code.

5. Results

No experiments have been conducted. This section will present observed exploratory findings, comparable model performance, uncertainty, error analysis, and interpretation. Tables and figures will be generated from recorded executions rather than illustrative or fabricated values.

6. Discussion and Limitations

The discussion will address selection bias, outcome completeness, missing data, prediction-time uncertainty, dependence between observations, generalization, and practical limits of the findings. The final scope must be consistent with the data and evaluation, without unsupported clinical or causal claims.

7. Conclusion

Conclusions are pending empirical results and joint author review. The final paper will answer the chosen research question using the observed evidence and state appropriate future work.

Use of Generative AI

ChatGPT/Codex, provided by OpenAI [OpenAI 2026], assisted with requirement review, source research, and this scaffold on October 12, 2026. The authors must verify scientific claims and follow Moodle citation guidance. Prompts, uses, verification, and limitations are recorded in supplementary material. Scientific author review remains pending.

References

Instituto Brasileiro de Geografia e Estatística (2022). Pesquisa nacional de saúde 2019: Microdados e documentação. https://ftp.ibge.gov.br/PNS/2019/Microdados/. Accessed October 12, 2026; candidate source, not yet selected.

Ministério da Saúde (2026). Banco de dados da síndrome respiratória aguda grave (srag) – 2019 a 2026. https://dadosabertos.saude.gov.br/dataset/srag-2019-a-2026. Accessed October 12, 2026; candidate source, not yet selected.

OpenAI (2026). Chatgpt. https://chatgpt.com/. AI assistance on October 12, 2026. Exact model version unavailable; prompts and limitations documented in supplementary material. Citation format pending Moodle guidance.